SIGNAL & DATA ANALYSIS IN NEUROSCIENCE 2020
STOCHASTIC PROCESS & FIRING RATE

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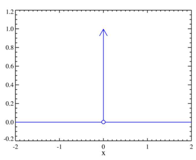
Outline

- Dirac's delta function
- Stochastic process
- Measuring firing rates
- Convolution

Dirac's delta function

- Dirac's δ function: $\delta(x-a) = 0 \text{ for } x \neq a, \quad \int_{-\infty}^{\infty} \delta(x)dx = 1$
- Kronecker's δ function: $\delta[i] = \delta_i = \begin{cases} 1 & i = 0 \\ 0 & \text{else} \end{cases}, \quad \delta_{ij} \begin{cases} 1 & i = j \\ 0 & \text{else} \end{cases}$
- Properties:

$$\int_{t=-\infty}^{\infty} X(t) \cdot \delta(\tau - t)dt \quad = \quad ?$$
$$\sum_{n=-\infty}^{\infty} X[n] \cdot \delta(k-n)dt \quad = \quad ?$$



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Dirac's delta function

- We can now express each spike (point process) with a certain delta function and then sum them to get continuous representation of the point process.

$$\rho(t) = \sum_{i=1}^{n} \delta(t - t_i)$$

- Why is it important? Mainly for analytic solutions.

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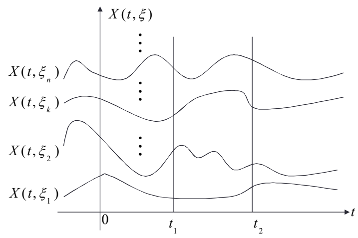
Stochastic process

2 types of averaging:

- Average over instances (trials)
- Average over time.



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Stochastic process

- **Stochastic process (**random process): A process whose evolution involves some indeterminacy described by probability distributions. In the simplest case the process amounts to a sequence of RV known as a time series.
- **Stationary process :** A stochastic process in which the pdf of some RV X does not change over time.
- **Weak Sense Stationary process (WSS)**: a process whose first two moments don't change over time:
  $E\{X(t)\} = \mu$ and $E\{X(t_1)X^*(t_2)\} = R_{xx}(t_1 - t_2),$
- **Ergodic process:** If averaging over time and sample space are equal the process is ergodic.
- An ergodic process is always stationary, the reverse may not be true.

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Stochastic process - examples

Determine whether the following examples represent a stationary /non-stationary /ergodic process:

- Drawing normal dice.
- Drawing biased dice.
- Gaussian noise: noise(t) ~N(μ,σ^2)
- Gaussian noise passed through a half-wave rectifier.
- A process X(t) = Y cos(ωt + θ), where
  ω is a constant, Y and θ are independent RVs.
  θ is uniformly distributed in [0, 2π].

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Measuring firing rate

- **There is no single standard definition for FR**
- r – rate over the whole period T also called *spike count rate:* nSpikes/T.
- <r> - rate averaged over all the trials, also called *average firing rate*
- *r(t)* – trial average rate over a short period (Δt→0).
  this method is sensitive to the Δt parameter and
  the smoothing method.

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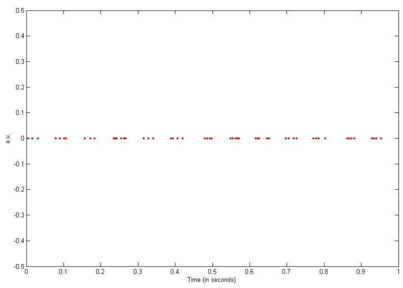
Firing rate - example



- Spike times

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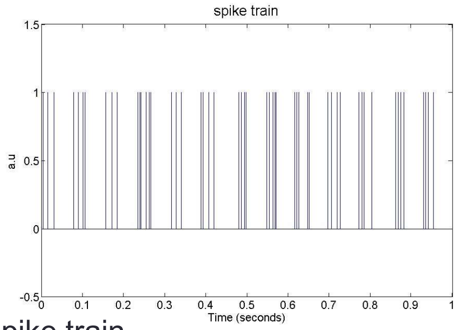

- Spike train

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Firing rate 1 – Total firing rate

- Number of Spikes/Total time
- = 53 spikes/second

- Nice – but we can do more.
- What if we divide the 1 second time span into several windows ?

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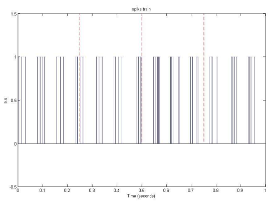
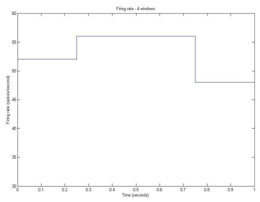
Firing rate 2 - Windowing

- 4 non-overlapping windows





- Better, but not enough

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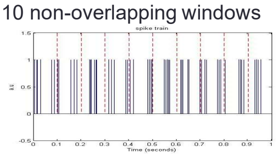
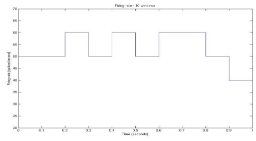
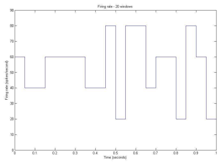
Firing rate

- 10 non-overlapping windows





- And even some more – 20 windows



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Smoothing data

- Data is generally noisy, you might need to apply a smoothing algorithm to expose its features.
- In general, smoothing algorithms work on windows. Windows are characterized by span & shape
- The **span** defines a window of neighboring points to include in the smoothing calculation for each data point.
- Large span – smoother graph with less resolution.
- Smaller span – “sharper” graph with higher resolution.
- The **shape** is the function that is applied to the points in the span.

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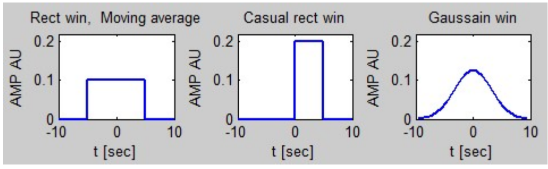
The smoothing window

- Common windows:



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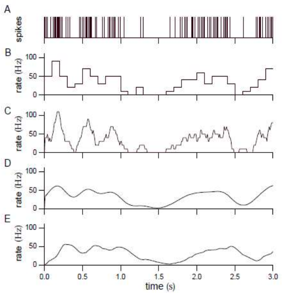
Firing rate – smoothing example



- A: Spikes
- B: Binned count
- C: Sliding window
- D: Sliding Gaussian kernel
- E: Sliding causal kernel

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Convolution

- Definition: the integral of the product of the two functions after one is reversed and shifted.
- Continuous conv: $Y(t) = (X * H)(t) = \int_{\tau=-\infty}^{\infty} X(\tau) \cdot H(t-\tau)d\tau$
- Discrete conv: $Y[m] = (X * H)[m] = \sum_{n=-\infty}^{\infty} X[n] \cdot H[m-n]$
- Matlab: Y = conv(X,H)

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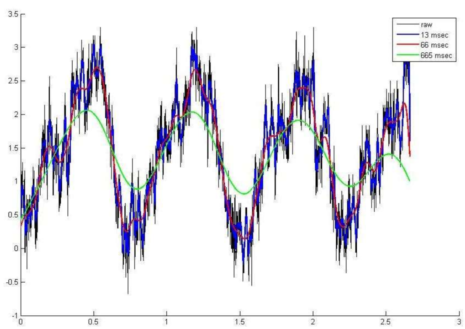
Example: convolution by a Gaussian window



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Demo